Cindy Zhang (1008902305)

Homework #4

**Lexical and Morphological Design Principles**

The lexicon of the language is designed to be mostly a priori, rather than derived directly from existing natural languages, though this decision remains open to revision pending further research into cross-linguistically common vocabulary. Some motivated phonological associations are nonetheless incorporated. For example, in many Indo-European languages, suffixes containing *-a* are associated with feminine meanings, while *-o* is associated with masculine meanings, reflecting widely attested sound-symbolic and grammatical gender tendencies across languages. Similarly, negation is marked with a nasal onset, aligning with a strong typological tendency for negative markers to contain nasal segments.

**Lexical Contrasts and Semantic Granularity**

Synonyms are deliberately constrained. Where lexical contrasts exist, they are typically realized as minimal pairs, sharing the same initial consonant and differing only in a single vowel feature. This design choice keeps phonological complexity low while still allowing systematic semantic contrasts.

For human kinship terms, especially sibling relations, the language allows optional distinctions for age (older vs. younger) and gender. Many languages obligatorily encode such distinctions (e.g., Hindi distinguishes older and younger siblings), while other languages, such as English, do not. To accommodate speakers of different native languages, the system is designed to be adaptable: speakers may choose how much semantic specificity to encode. For example, a Hindi speaker may consistently mark older/younger distinctions, while an English speaker may omit them without loss of grammaticality.

This flexibility aligns with a broader design goal: the language should be usable by speakers from diverse linguistic backgrounds, allowing optional modifiers to increase semantic precision when desired, in a manner reminiscent of philosophical or engineered languages but without enforcing maximal specificity.

**Morphological Structure and Word Classes**

At present, all affixes are suffixes, yielding a consistently head-initial structure. The core semantic content of a word is presented first, with additional specification added later. This

mirrors analytic tendencies in many natural languages and simplifies parsing, particularly for computational applications.

Another notable feature is that the language adopts flexible word classes. Lexemes are not inherently marked as nouns, verbs, or adjectives; instead, their syntactic role is determined entirely by word order or syntactic position. This approach is common cross-linguistically, particularly in analytic languages (e.g., Mandarin Chinese) and has also been implemented in constructed languages such as *toki pona*. While this flexibility may pose challenges for NLP systems, the language compensates through fixed word order and highly regular syntax. Optionally, a part-of-speech morpheme could be introduced if explicit disambiguation is required (will be added later).

**Phonological Economy and Frequency**

The lexicon does not strictly follow a Zipfian frequency distribution, primarily because the phonological inventory is intentionally constrained: there are only about 40 possible syllables. To preserve usability, highly frequent or semantically central words are preferentially assigned monosyllabic forms. Syllables beginning with common consonants are reused for morphemes with broad or abstract meanings, maximizing combinatorial potential

Some lexemes also function as affixes, further increasing morphological efficiency and reducing the total number of primitive forms learners must acquire.

**Conceptual Update: Underspecification, Complexity, and QNLP**

An update to the overall design reframes the language as one that deliberately underspecifies lexical and grammatical meaning until additional function words or morphemes are introduced. This approach is inspired both by quantum mechanics and by emerging work in quantum natural language processing (QNLP), as well as by the development of pidgins and creoles.

In many pidgins and early creole stages, learners first acquire highly simplified structures sufficient for minimally viable communication, with complexity added gradually and selectively. Similarly, in this language, complexity is not embedded directly in core lexical items but is instead introduced via function words that act like operators. This makes semantic enrichment explicit and modular, and it also facilitates machine parsing and decoding.

Polysemy is treated not as a flaw to be eliminated but as an inherent and useful property of language. Rather than enforcing rigid one-to-one mappings between form and meaning, as many philosophical languages have attempted, the language represents polysemy as a set of probability amplitudes, where some senses are more likely or frequent than others. This will become more apparent once the lexicon is expanded beyond basic concepts.

The lexicon therefore leaves room for potential morphemes encoding likelihood, uncertainty, and evidentiality. These morphemes might interact in interesting ways with determiners: indefinite determiners correspond to semantic states prior to “collapse” (i.e., before reference is fixed), while definite determiners correspond to post-collapse states in which reference has been resolved. These will be interesting directions to explore in later stages of building the language.

**The lexicon**

Note: some words are AI-generated after inputting the principles and constraints above. I wrote the constraints before attempting to generate the words, and realized that it was very difficult to stick to them in practice, so I could not complete the word list in time.

| Meaning | Lexeme | Notes |
|---|---|---|
| **animals** | | |
| animal | ki | |
| bird | kito | Animal+sky |
| worm | kisalati | Animal+earth+long+small |
| fish | kilo | Animal+water |
| snake | kisala | animal+earth+long |
| dog | kisa | Animal+earth |
| **adjectives** | | |
| black | mimo | |
| white | pemi | |
| red | meti | |
| yellow | nimo | |
| grue (green/blue) | nale | |
| good | so | |
| bad | se | |
| old | la | |
| young / new | le | |
| cold | mile | |
| warm / hot | mulo | |
| wet | mahu | |
| dry | mahi | |
| big | ta | |

| Meaning | Lexeme | Notes |
|---|---|---|
| small | ti | |
| thick | musi | |
| thin | mose | |
| long / tall | hako | |
| short | haku | |
| sharp | nesu | |
| dull | naso | |
| light (weight) | kimi | |
| heavy | kumu | |
| rotten | kino | |
| fresh | kenu | |
| **humans** | | |
| person | ke | Same word as who |
| man | kepo | Person+male |
| woman | kepa | Person+female |
| child | keti | Person+small |
| spouse | kelo | |
| husband | kelopo | Spouse+male |
| wife | kelopa | Spouse+female |
| parent | kela | |
| father | kelapo | Parent+man |
| mother | kelapa | Parent+woman |
| sibling | keni | |

| Meaning | Lexeme | Notes |
|---|---|---|
| brother | kenipo | Sibling+male |
| Older brother | kenipola | Sibling+male+old |
| Younger brother | kenipole | Sibling+male+young |
| sister | kenipa | Sibling+male |
| Older sister | kenipala | Sibling+female+old |
| Younger sister | kenipale | Sibling+female+young |
| **animal parts** | | |
| head | pa | |
| hair | pasa | |
| eye | | |
| nose | | |
| mouth | | |
| tooth | | |
| tongue | | |
| ear | | |
| neck | | |
| heart | | |
| arm | | |
| hand | | |
| belly | | |
| leg | | |
| foot | | |
| skin | | |
| blood | | |
| bone | | |
| feather | | |
| horn | | |
| claw | | |
| wing | | |
| tail | | |
| **other nouns** | | |

| Meaning | Lexeme | Notes |
|---|---|---|
| night | | |
| day | | |
| year | | |
| sky | | |
| cloud | | |
| dirt / earth | | |
| sun | | |
| moon | | |
| star | | |
| grass | | |
| tree | | |
| stick | | |
| bark | | |
| leaf | | |
| flower | | |
| fruit | | |
| stone | | |
| mountain | | Stone+big |
| wind | | |
| smoke | | |
| fire | | |
| ashes | | |
| water | | |
| ice | | First syllable is water |
| snow | | First syllable is water |
| rain | | First syllable is water |
| sea | | First syllable is water |
| lake | | First syllable is water |
| river | | First syllable is water |

| Meaning | Lexeme | Notes |
|---|---|---|
| seed | | First syllable is water |
| egg | | |
| salt | | |
| rope | | |
| road | | |
| sand | | |
| **verbs** | | |
| think | | |
| know | | |
| see | | Same word as eyes |
| smell | | Same word as nose |
| hear | | Same word as ear |
| say | | Same word as mouth |
| give | | |
| take | | |
| live | | |
| die | | |
| eat | | |
| drink | | Eat+water |
| breathe | | |
| laugh | | |
| blow | | |
| sing | | |
| work | | |
| play | | |
| dance | | |
| cook | | |
| sew | | |
| swim | | |
| walk | | |
| fall | | |

| Meaning | Lexeme | Notes |
|---|---|---|
| sit | | |
| stand | | |
| sleep | | |
| push | | |
| pull | | |
| dig | | |
| cut | | |
| throw | | |
| scratch | | |
| bite | | |
| kill | | |
| **personal pronouns** | | |
| I | mi | |
| we | mino | I+many |
| you (SG) | mu | |
| you (PL) | muno | You (SG)+many |
| Gender neutral third person singular | me | |
| he | mepo | 3rd person+masculine suffix |
| she | mepa | 3rd person+feminine suffix |
| it | me | |
| they | meno | 3rd person+many |
| **miscellaneous** | | |
| what | ki | |
| when | ku | |
| where | ka | |
| who | ke | |
| how | ko | |
| this | li | |

| Meaning | Lexeme | Notes |
|---|---|---|
| that | lu | |
| there | lika | This+where |
| here | luka | That+where |
| and | hi | |
| if | hu | |
| because | ho | |
| with | he | |
| not | mo | |
| all | na | |
| few | ni | Some iconicism – the vowel in *few* is more closed than the vowel in *some* and *all* |
| some | ne | |
| many | no | |
| none | nu | |
| **language-specific words** | | |
| learn | hita | Speakers of this language will be users of a language learning app |

| Meaning | Lexeme | Notes |
|---|---|---|
| video | pota | |
| home | lami | This can be used to talk about where one is from or what native language(s) they speak |
| App (as in software) | mola | Phonetically similar to the language learning app Mora |
| language | | Speak/mouth+home |
| computer | kaputa | Loanword (since many natlangs have loanwords for technology) |
| phone | kaputati | Computer + small |
| exchange | nelo | |
| country | tepa | |
| repeat | lala | Some iconicism through reduplication |